

Multi-frame coherent track-before-detect method for weak tones in passive sonar

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ABSTRACT

To address the performance degradation caused by power fluctuations of tones, the multi-frame coherent integration-based track-before-detect (TBD) method is proposed. In the proposed method, the high gain advantage of coherent integration and the information accumulation ability of multi-frame TBD method based on batch processing technique are combined to improve the detection and tracking performance of fluctuating tones. Firstly, a new measurement model is established with batch processing technology and coherent integration across multiple data frames is achieved utilizing the tonal coherence, thus the SNR in measurement is greatly improved. Then, a new likelihood ratio function is derived to match the established measurement model, ensuring that TBD processing can be performed correctly. Finally, the effects of the newly established likelihood ratio and the batch length on the detection performance are analyzed in detail through theoretical derivations. The superiorities and robustness of the proposed method are demonstrated through simulation analysis and processing results for sea experimental data.

1. Introduction

Passive sonar accomplishes missions such as detection, positioning, tracking and recognition by processing underwater noise radiated by the target [1–3]. Research has shown that narrowband components (known as tones) in the radiated noise of underwater vehicles and surface vessels are more energetic compared to broadband continuous spectra [4,5]. Therefore, it is often used as an important feature for target detection in passive sonar [6]. Currently, there are still many difficulties in tone detection. Firstly, the power of the tone is dispersed across multiple discrete frequency bins because of the multipath interference and the relative motion between the source and receiver. This not only reduces the SNR in each frequency bin, but also weakens the concentration of narrowband tones in the frequency domain. Moreover, in complex underwater acoustic channels, the fluctuations in target amplitude can cause unstable SNR of received signal. When the SNR is high, the target can be easily detected. But when the SNR is low, the detector will lose the target or even pursue the wrong target. Therefore, it is necessary to develop reliable detectors that can robustly detect tones with frequency variation and intensity fluctuation in complex marine environments.

The classical power spectrum estimation methods based on the discrete Fourier transform (DFT), has been widely used to detect tones with constant frequency in wideband noise. These methods fall into two main categories. The first category is non-coherent methods [7–10]. The proportion of noise and interference in the signal is reduced by means of linear or non-linear averaging over multiple data segments. However, the phase information of the signal is ignored during this process. When the target SNR is low, the performance may be reduced rapidly. The second category is coherent methods [11–14]. By compensating the phase difference between data segments, coherent integration can be achieved for signals of interest in each segment. Instead, the phase of noise is random and cannot be coherently integrated. Therefore, higher signal processing gains can be achieved through coherent integration methods, which is more conducive to signal detection. In [12], coherent accumulation was achieved by taking the maximum of the DFT results for the segments in each frequency bin. Compared to the periodogram, the variance is greatly reduced owing to the operation of taking maximum. More importantly, the sensitivity of the estimated spectrum to Doppler frequency shift is decreased. It provides a new solution for coherent processing.

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Time-frequency analysis is a powerful tool for detecting narrowband signals with Doppler frequency shift. Generally, the tones are represented as time-varying tracks in time-frequency domain by the low frequency analysis and recording (LOFAR) technique [15]. A number of methods have been proposed to detect tonal tracks in LOFAR. Streit et al. [16] design a complete processing procedure for tone detection using Hidden Markov Models (HMM). The HMM tracker operating on the thresholded output of LOFAR demonstrates satisfactory performance in different synthetic datasets. Subsequently, Paris et al. modeled the output trajectory of [16] as a first order Markov chain and automatically determined the tonal initiation and termination by executing the HMM tracker again [17]. Later, they proposed a non-parametric likelihood function building method based on the normalized probabilistic integration of the spectral power, allowing detection in the absence of a priori SNR [18]. But the performance of the HMM tracker suffers a slight loss. Xie et al. [19,20] developed a multitrack tracking algorithm to provide a solution for the multi-tone tracking problem. However, it requires a further separation of the output mixed tracks to obtain each individual tone track. Although the application of HMM in tonal detection and tracking has been further developed [21,22] there is still a lack of a comprehensive and efficient solution to the above-mentioned problems.

In recent years, a particle filter-based track-before-detect (PF-TBD) method has been widely used for weak target detection [23–27]. Non-thresholded original data rather than thresholded result is directly used as the inputs of the tracker, thus avoiding the miss of weak targets. In passive sonar, the PF-TBD method has shown excellent performance in underwater source detection [28], tracking [29], and localization [30]. In addition, it is also used to track the azimuth of unknown targets in the bearing-time record spectrum [31,32]. The key problem in TBD is how to design an effective likelihood function. Ref. [31] designed a data fitting scheme based on the Cramer-von distance criterion to determine the optimal likelihood function form. In [32], the likelihood function is directly defined as the accumulation of raster bin values from the spectral measurement associated with the past state estimates of each particle. Recently, Zhang et al. [33] developed a soft-decision method based on PF-TBD to detect weak tones in LOFAR. A strictly analytical expression for likelihood ratio function is derived to match the statistical characteristics of the amplitude spectrum in LOFAR. The percentage of surviving particles is defined as the existence probabilities of tones. In this way, the initiation and termination of tones can be flexibly estimated. But this method only utilizes measurement information from a single frame, which may result in discontinuous and unstable detection results for tones with power fluctuations.

The multi-frame track-before-detect (MF-TBD) method provides an effective solution to detect fluctuating targets [34,35]. In MF-TBD, the signal energy of multiple consecutive moments (which can be referred to as frames) is incoherently integrated by a sliding-window-based batch processing technique [36]. Therefore, the target SNR to be detected is improved, allowing the processor to attenuate noise-related detection and amplify target-related detection. MF-TBD methods have achieved remarkable performance in weak target detection for radar and sonar [34–39]. However, most of the existing MF-TBD methods are based on dynamic programming algorithm. When tracking multiple targets, it is necessary to have a prior knowledge of the target number. Furthermore, non-coherent integration cannot provide enough gain in very low SNR situations, resulting in a severe loss of detection performance in existing MF-TBD methods. As mentioned earlier, coherent integration can further improve the target SNR compared with non-coherent integration. Therefore, it is reasonable to believe that if the phase information between consecutive frames is fully utilized, the detectability of the target under low SNR can be further enhanced.

Our work aims to achieve more robust detection performance for targets with slow frequency variation and large fluctuations in target power. With this goal, a multi-frame TBD method originated from the extension of the single frame TBD (referred to as SF-TBD hereafter)

method in [33] is designed and combined with the coherent integration processing in [12]. For the combination, the main issues are how to design an effective likelihood ratio function and select the appropriate number of frames for a batch. In this study, a novel measurement model is established firstly with batch processing based on the coherent accumulation theory of multiple data frames. A rigorous mathematical derivation is then developed to obtain the analytical expression of the likelihood function, enabling it to match the statistical characteristics of multi-frame measurements accurately. Theoretical analysis indicates that the proposed method can obtain more additional gain as the batch length increases. Finally, the theoretical derivations and simulations on the performance of the proposed method are conducted to provide a reference for the selection of batch length. The simulation results show that the proposed algorithm can improve the target SNR, thereby obtaining a higher detection probability than the SF-TBD algorithm. Different trial data processing results also verify the advantages and robustness of the proposed algorithm, even when the target exhibits a large fluctuation.

The remainder of this study is organized as follows. Section II introduces the signal model and single-frame detection scheme. The principle and implementation steps of the proposed multi-frame coherent TBD (referred to as MFC-TBD) method, as well as a comparative analysis with the SF-TBD method, are presented in Section III. Numerical simulation and experimental data processing are presented in Section IV and Section V, respectively. Section VI summarizes the conclusion.

2. Problem formulation

2.1. Signal model

Consider that the received signal in passive sonar consists of a pure tone and background noise. The N – point discrete time series received can be expressed as

$$x(n) = s(n) + g(n) = A(n)\cos[2\pi f_0(n)n] + g(n), \quad n = 0, 1, \dots, N-1 \quad (1)$$

Where $A(n)$ and $f_0(n)$ represent the time-varying amplitude and frequency of the tone, respectively. $g(n)$ is an additive Gaussian white noise with zero mean and variance σ^2 .

In order to better analyze the time-varying characteristics of received signal in (1), it is necessary to transform it to the time-frequency domain, as shown in Fig. 1. The original signal in Fig. 1(a) is split into K segments with M samples each. Assuming that the amplitude and frequency of the tone is constant in each segment, the k^{th} signal segment in Fig. 1(b) can be written as

$$x_k(n) = s_k(n) + g_k(n) = A_k \cos[2\pi f_{0k}(n + kM)] + g_k(n) \quad n = 0, 1, \dots, M-1; k = 0, 1, \dots, K-1 \quad (2)$$

Where A_k and f_{0k} represent the amplitude and frequency of the tone in k^{th} segment. $g_k(n)$ is the k^{th} noise segment.

Performing M – point DFT on single data segment, the spectral results of each signal segment are shown in Fig. 1(c). By only considering positive frequency values, the DFT results of k^{th} signal segment are written as

$$\begin{aligned} Y_k(l) &= \frac{1}{M} \sum_{n=0}^{M-1} s_k(n) \exp\left\{-j2\pi \frac{l}{M} n\right\} + \frac{1}{M} \sum_{n=0}^{M-1} g_k(n) \exp\left\{-j2\pi \frac{l}{M} n\right\} \\ &= \frac{A_k}{2M} \exp\{j2\pi k f_{0k} M\} \exp\{j\pi(M-1)(f_{0k}M - l)\} \\ &\quad / \left\{ \frac{\sin[\pi(f_{0k}M - l)]}{\sin[\pi(f_{0k}M - l)/M]} + G_k(l) \right\} \end{aligned} \quad (3)$$

Where l is the index of discrete frequency bins. $G_k(l)$ represents the DFT results of k^{th} noise segment $g_k(n)$. As shown in (3), the spectrum of each signal segment is a complex sequence, including amplitude and

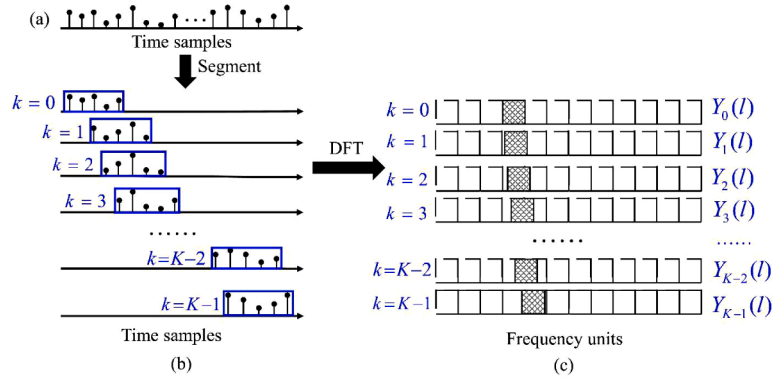


Fig. 1. The process of transforming a signal from time domain to time-frequency domain. (a) Discrete signal received. (b) Segmenting the signal. (c) Spectral results of signal segments.

phase. In the SF-TBD method, only amplitude is sent to the subsequent TBD filter, while the phase information is ignored.

2.2. SF-TBD method

A soft-decision method based on PF-TBD to detect weak tones in LOFAR is developed in [33]. Given that this method only utilizes DFT spectral information from a single frame, we called it SF-TBD (single frame TBD) method. Its basic principle is re-introduced.

2.2.1. Dynamic model

As shown in (2), the target state to be estimated includes the tonal amplitude and frequency, i.e. the state vector can be defined as $X_k = [f_{0k}, A_k]^T$ at time step k . Then the dynamic model of state evolution is represented as:

$$X_{k+1} = X_k + V_k \quad (4)$$

Where $V_k = [q_{f_0}, q_A]^T$, $k = 0, 1, \dots, K-1$, q_{f_0} and q_A are the random process noise for the frequency and amplitude, respectively.

Generally, the tonal-existence indicator variable e_k is defined to describe the target detection problem in the TBD filter [24,33]. It is modelled as a two-state Markov chain with a tone either existing in the data, $e_k = 1$ or absent from the data, $e_k = 0$. The probability transition matrix can be written as

$$\Pi = \begin{bmatrix} 1 - P_b & P_b \\ P_d & 1 - P_d \end{bmatrix} \quad (5)$$

where P_b represents the tonal-birth probability and P_d represents the tonal-death probability. They are respectively defined as

$$P_b \triangleq P\{e_k = 1 | e_{k-1} = 0\} \quad (6)$$

$$P_d \triangleq P\{e_k = 0 | e_{k-1} = 1\} \quad (7)$$

2.2.2. Measurement model and likelihood ratio function

The measurement Z_k at time k in SF-TBD method can be described as

$$Z_k = \begin{cases} h(X_k, g_k) & e_k = 1 \\ h(g_k) & e_k = 0 \end{cases} \quad (8)$$

Where the certain function $h(\cdot)$ is related to the DFT process in Fig. 1. In (8), g_k is a random measurement noise. If a tone exists at time k , the measurement is entirely determined by tone and random noise. Conversely, if the tone is absent at time k , the measurement only reflects the characteristics of the noise.

As shown in Fig. 2, spectral result of one signal segment is considered as one frame. The measurement at time k is defined as the module of the spectrum in k^{th} frame,

$$Z_k^s(l) = |Y_k(l)| \quad (9)$$

Define the DFT bin closest to the true tonal frequency f_{0k} is l_{0k} . With the distribution of $g_k(n)$, we can know that $Y_k(l)$ follows a complex Gaussian distribution. For the noise bins, i.e., $l \neq l_{0k}$, we have

$$Y_k(l) \sim CN(0, 2\sigma_G^2) \quad (10)$$

and for the tonal bin, i.e., $l = l_{0k}$, we have

$$Y_k(l) \sim CN(S_k(l), 2\sigma_G^2) \quad (11)$$

where $S_k(l)$ and σ_G^2 are represented as

$$S_k(l) = \frac{A_k}{2M} \exp\{j2\pi k f_{0k} M\} \exp\{j\pi(M-1)(f_{0k}M - l)/M\} \frac{\sin[\pi(f_{0k}M - l)]}{\sin[\pi(f_{0k}M - l)/M]} \quad (12)$$

and

$$\sigma_G^2 = \frac{\sigma^2}{2M} \quad (13)$$

Then the distribution of the measurement $Z_k^s(l)$ when $e_k = 0$ is

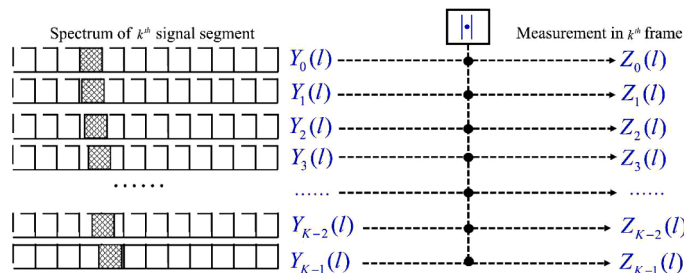


Fig. 2. The measurements of SF-TBD method.

$$p(Z_k^s(l)|e_k=0) = \frac{Z_k^s(l)}{\sigma_G^2} \exp\left\{-\frac{Z_k^s(l)^2}{\sigma_G^2}\right\} \quad (14)$$

and when $e_k = 1$ is

$$p(Z_k^s(l)|e_k=1) = \frac{Z_k^s(l)}{\sigma_G^2} \exp\left\{-\frac{|S_k(l)|^2 + Z_k^s(l)^2}{2\sigma_G^2}\right\} I_0\left(\frac{|S_k(l)|Z_k^s(l)}{\sigma_G^2}\right) \quad (15)$$

In terms of (14)-(15), the likelihood ratio function for SF-TBD method can be constructed as

$$l(Z_k^s(l)|X_k, e_k) = \frac{p(Z_k^s(l)|X_k, e_k=1)}{p(Z_k^s(l)|X_k, e_k=0)} = \exp\left\{-\frac{|S_k(l)|^2}{2\sigma_G^2}\right\} I_0\left(\frac{|S_k(l)|Z_k^s(l)}{\sigma_G^2}\right) \quad (16)$$

At this point, the SF-TBD method can be successfully implemented. However, single-frame detection will cause intolerable missed alarms of the target in the presence of acoustic signal fluctuation. To overcome this problem, we combine coherent information between multiple frames to develop a novel TBD model.

3. Proposed MFC-TBD method

To improve the detection and tracking performance of tones, the new measurement model and likelihood ratio function based on batch processing technology and multi-frame coherent integration are first established. The process of implementing recursive TBD through particle filter is then presented. Finally, a detailed analysis is conducted on the advantages of the proposed MFC-TBD method compared to the SF-TBD method.

3.1. Measurement model and likelihood ratio function

In MFC-TBD method, the information from W frames is fused and formed into a batch. As shown in Fig. 3, the batch moves forward one step each time and the batch length is defined as W . Unlike the SF-TBD method, which only utilizes the amplitude spectrum of single data frame, in MFC-TBD method, not only the amplitude spectra of multiple data frames are utilized but also the coherent phase between multiple frames are considered. The first task of the proposed method is to achieve coherent integration during W data frames.

Denote the κ^{th} batch by $B_\kappa(l)$, $\kappa = 0, \dots, K - W$. Define the tonal frequency of the κ^{th} batch as $f_{0\kappa}$. The spectra included in each batch are expressed as

$$B_\kappa(l) = \{Y_\kappa(l), \dots, Y_{\kappa+W-1}(l)\} \quad (17)$$

In practice, the tonal frequency always changes with time due to source motion or sound field perturbations. Define the change rate of tonal frequency as $\dot{f}(t)$. If the condition

is satisfied, the tonal frequency can be approximated as a constant in a single batch. Where $\Delta f = \frac{1}{T_0}$ is the width of discrete frequency bin and T_0 is the time length of a single frame. In this way, the phase difference between adjacent frames in κ^{th} batch $B_\kappa(l)$ can be approximated as a complex sinusoidal sequence. It can be compensated by taking a W -point DFT (padding zeros if necessary) on $B_\kappa(l)$ [12]:

$$\begin{aligned} Q_{B_\kappa}(l, q) &= \frac{1}{W} \sum_{i=0}^{W-1} Y_{\kappa+i}(l) \exp\left(-j2\pi q \frac{i}{W}\right) \\ &= Q_{S_\kappa}(l, q) + Q_{G_\kappa}(l, q) \end{aligned} \quad (19)$$

where $Q_{S_\kappa}(l, q)$ and $Q_{G_\kappa}(l, q)$ are represented as follows

$$\begin{aligned} Q_{S_\kappa}(l, q) &= \frac{A_\kappa}{2WM} \exp\{j\pi(f_{0\kappa}M - l)(M - 1)/M\} \frac{\sin[\pi(f_{0\kappa}M - l)]}{\sin[\pi(f_{0\kappa}M - l)/M]} \\ &\quad \exp\{j\pi(Wf_{0\kappa}M - Wl - q)(W - 1)/W\} \frac{\sin[\pi(Wf_{0\kappa}M - Wl - q)]}{\sin[\pi(Wf_{0\kappa}M - Wl - q)/W]} \end{aligned} \quad (20)$$

$$Q_{G_\kappa}(l, q) = \frac{1}{W} \sum_{i=0}^{W-1} G_{\kappa+i}(l) \exp\left(-j2\pi q \frac{i}{W}\right) \quad (21)$$

For each discrete frequency bin l , the DFT bin q_l corresponding to complex sinusoidal sequence hidden in the phase difference is given as

$$q_l = \underset{q}{\operatorname{argmax}} [|Q_{B_\kappa}(l, q)|] \quad (22)$$

For the tonal bin $l_{0\kappa}$, the value of term $\frac{\sin[\pi(f_{0\kappa}M - l)]}{M \sin[\pi(f_{0\kappa}M - l)/M]}$ in (20) equals to 1, which is greater than the corresponding value for all noise bins. This implies that by compensating the phase difference in different data frames, coherent integration can be efficiently achieved. The noise phase, instead, is different in each frame and synchronous accumulation cannot be achieved. As a result, the target SNR can be greatly improved. Therefore, the measurement model for MFC-TBD method is denoted as

$$Z_\kappa^M(l) = \begin{cases} h(X_\kappa, g_\kappa) = |Q_{B_\kappa}(l_{0\kappa}, q_{l_{0\kappa}})|^2 & e_\kappa = 1 \\ h(g_\kappa) = |Q_{B_\kappa}(l, q_l)|^2 & l \neq l_{0\kappa}, e_\kappa = 0 \end{cases} \quad (23)$$

where the nonlinear function $h(\cdot)$ is determined by the multi-frame coherent integration process. Then the next task of the proposed method is to obtain the likelihood ratio function.

With the distribution of $Y_\kappa(l)$, the distributions of $Z_\kappa^M(l)$ when $e_\kappa = 0$ and $e_\kappa = 1$ can be expressed as follows (The detailed derivation is given in Appendix), respectively.

$$p(Z_\kappa^M(l)|e_\kappa=0) = \frac{W}{2\sigma_{Q_G}^2} \left[1 - \exp\left\{-\frac{Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\}\right]^{W-1} \exp\left\{-\frac{Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\} \quad (24)$$

$$\begin{aligned} p(Z_\kappa^M(l)|e_\kappa=1) &= \left[1 - \exp\left\{-\frac{Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\}\right]^{W-1} \frac{1}{2\sigma_{Q_G}^2} I_0\left(\frac{\sqrt{|Q_{S_\kappa}(l, q_l)|^2 Z_\kappa^M(l)}}{\sigma_{Q_G}^2}\right) \exp\left\{-\frac{|Q_{S_\kappa}(l, q_l)|^2 + Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\} \\ &\quad + \frac{(W-1)}{2\sigma_{Q_G}^2} \left[1 - \exp\left\{-\frac{Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\}\right]^{W-2} \exp\left\{-\frac{Z_\kappa^M(l)}{2\sigma_{Q_G}^2}\right\} \int_0^{Z_\kappa(l_0)} P_{\tilde{Z}_\kappa(l_{0\kappa}, q_{l_{0\kappa}})}(x) dx \end{aligned} \quad (25)$$

In light of (24) and (25), we can get the likelihood ratio (LR):

$$\left| \int_{t_\kappa}^{t_\kappa + WT_0} \dot{f}(t) dt \right| < \Delta f \quad (18)$$

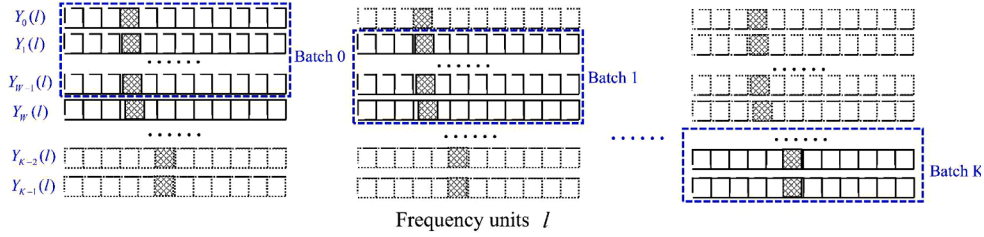


Fig. 3. Illustration of batch processing in MFC-TBD method.

$$\begin{aligned}
 l(Z_k^M(l)|X_k, e_k) &= \frac{p(Z_k^M(l)|X_k, e_k = 1)}{p(Z_k^M(l)|X_k, e_k = 0)} \\
 &= \frac{1}{W} I_0 \left(\frac{\sqrt{|Q_{S_k}(l, q_l)|^2 Z_k^M(l)}}{\sigma_{Q_G}^2} \right) \exp \left\{ -\frac{|Q_{S_k}(l, q_l)|^2}{2\sigma_{Q_G}^2} \right\} \\
 &\quad + \frac{W-1}{W} \left[1 - \exp \left\{ -\frac{Z_k^M(l)}{2\sigma_{Q_G}^2} \right\} \right]^{-1} \int_0^{Z_k(l)} P_{\tilde{Z}_k(l_{0k}, q_{l_{0k}})}(x) dx
 \end{aligned} \quad (26)$$

$$\hat{\sigma}_G^2 = \frac{1}{2W(M-1)} \sum_{i=0}^{W-1} \left[\sum_{l=0}^{l_{0k}-1} |Y_{k+i}(l)|^2 + \sum_{l=l_{0k}+1}^M |Y_{k+i}(l)|^2 \right] \quad (28)$$

For the multi-tone situations in practice, the protection window and reference window in [33] are used to reduce the estimation error of noise variance. Specifically, defining M_1 is the length of the protection window and M_2 is the length of the reference window, the noise variance σ_G^2 at time k is estimated by:

$$\hat{\sigma}_G^2 = \begin{cases} \frac{1}{2WM_2} \sum_{i=0}^{W-1} \left[\sum_{l=l_{0k}+M_1+1}^{l_{0k}+M_1+M_2} |Y_{k+i}(l)|^2 \right] & 0 \leq l_{0k} \leq M_1 \\ \frac{1}{2W(M_2 - M_1 + l_{0k})} \sum_{i=0}^{W-1} \left[\sum_{l=0}^{l_{0k}-M_1-1} |Y_{k+i}(l)|^2 + \sum_{l=l_{0k}+M_1+1}^{l_{0k}+M_1+M_2} |Y_{k+i}(l)|^2 \right] & M_1 < l_{0k} \leq M_1 + M_2 \\ \frac{1}{4WM_2} \sum_{i=0}^{W-1} \left[\sum_{l=l_{0k}-M_1-M_2}^{l_{0k}-M_1-1} |Y_{k+i}(l)|^2 + \sum_{l=l_{0k}+M_1+1}^{l_{0k}+M_1+M_2} |Y_{k+i}(l)|^2 \right] & M_1 + M_2 < l_{0k} \leq M-1-M_1-M_2 \\ \frac{1}{2W(M_2 - M_1 + M - l_{0k} + 1)} \sum_{i=0}^{W-1} \left[\sum_{l=l_{0k}-M_1-M_2}^{l_{0k}-M_1-1} |Y_{k+i}(l)|^2 + \sum_{l=l_{0k}+M_1+1}^{M-1} |Y_{k+i}(l)|^2 \right] & M-1-M_1-M_2 < l_{0k} \leq M-2-M_1 \\ \frac{1}{2WM_2} \sum_{i=0}^{W-1} \left[\sum_{l=l_{0k}-M_1-M_2}^{l_{0k}-M_1-1} |Y_{k+i}(l)|^2 \right] & M-2-M_1 < l_{0k} \leq M-1 \end{cases} \quad (29)$$

The expression of $p_{\tilde{Z}_k(l_{0k}, q_{l_{0k}})}(x)$ is given in (51). The noise variance $\sigma_{Q_G}^2$ is given as

$$\sigma_{Q_G}^2 = \frac{\sigma_G^2}{W} \quad (27)$$

It is worth noting that the noise variance $\sigma_{Q_G}^2$ in (27) is unknown and time-varying in practical situations. The estimates of the noise variance are given in [9,33] by assuming that noise is stationary in time and frequency domain, respectively. For the former method, one needs to collect enough data frames in each batch to ensure an accurate estimate of noise variance. However, the signal coherence time will decrease as ocean fluctuations become more violent, which tends to greatly limit the available batch length. Through experiments, we find that latter method has more robust estimation performance in complex marine environments. Therefore, the maximum likelihood estimation of the σ_G^2 is given as

Finally, the maximum likelihood estimation of the $\sigma_{Q_G}^2$ can be obtained according to (27).

3.2. Implementation by particle filter

In this section, the particle filter (PF) method is used to achieve recursive TBD through Monte Carlo simulation. In PF method, a set of random samples (particles) with associated weights is generated to approximate the posterior distribution and the state estimates are computed based on these particles and weights. The PF method approaches the optimal Bayesian estimate as particles number becomes very large. On account of the general particle filter framework [23], the steps for implementing the MFC-TBD algorithm using PF method are presented in Algorithm 1.

It is well known that the tracking results of PF method are very

Algorithm 1

Implementation of MFC-TBD method.

-
1. Generate N_p particles, draw the prior state of i th particle $X_0^i \sim p_0(X_0)$, assign the prior weight, $w_0^i = \frac{1}{N_p}$.
 2. For $\kappa = 1 : K - W$
 3. Calculate the measurement at time κ according to (23);
 4. For $i = 1 : N_p$
 5. if a newly born particle, draw $X_\kappa^i \sim p(X_\kappa | e_\kappa = 1, e_{\kappa-1} = 0, Z_\kappa)$ (where $p(X_\kappa | e_\kappa = 1, e_{\kappa-1} = 0, Z_\kappa)$ follows the initial distribution $p_0(X_0)$);
 6. if a continuing particle, draw $X_\kappa^i \sim p(X_\kappa | X_{\kappa-1}^i, e_\kappa = 1, e_{\kappa-1} = 1, Z_\kappa)$ (where $p(X_\kappa | X_{\kappa-1}^i, e_\kappa = 1, e_{\kappa-1} = 1, Z_\kappa)$ is obtained from the dynamic mode (4));
 7. Using the likelihood ratio function in (26) to calculate the normalized weight of i th particle, $w_\kappa^i = \frac{w_{\kappa-1}^i l(Z_\kappa^M(l) | X_\kappa^i, e_\kappa)}{\sum_{j=1}^{N_p} w_{\kappa-1}^j l(Z_\kappa^M(l) | X_\kappa^j, e_\kappa)}$.
 8. End
 9. Resample the particles. Then all particles will have a uniform weight $\frac{1}{N_p}$ and the state of the i th particle is reassigned as $(\hat{X}_\kappa^i, \hat{e}_\kappa^i)$.
 10. Estimate tonal state $\hat{X}_\kappa$ and tonal-existence probability $P_{e,\kappa}$ at time κ from the particles with $\hat{e}_\kappa^i = 1$, $\hat{X}_\kappa = \frac{1}{\sum_{i=1}^{N_p} \hat{e}_\kappa^i} \sum_{i=1}^{N_p} \hat{X}_\kappa^i \hat{e}_\kappa^i$, $P_{e,\kappa} = \frac{\sum_{i=1}^{N_p} \hat{e}_\kappa^i}{N_p}$.
 11. End
-

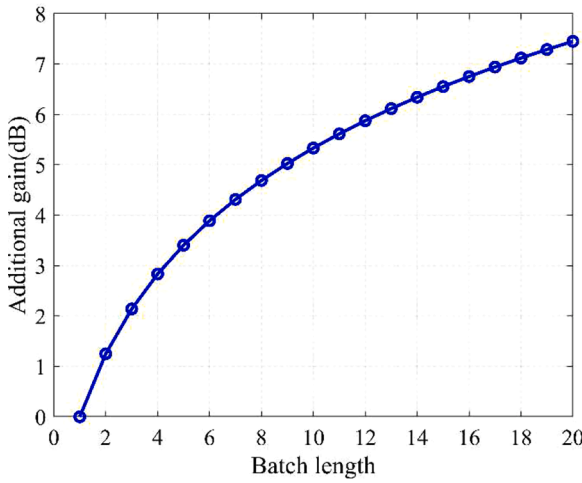


Fig. 4. Extra gain obtained by MFC-TBD compared with SF-TBD.

Table 1

Running time of the SF-TBD and MFC-TBD methods.

Instruction	SF-TBD (s)	MFC-TBD (s)
Step1: Calculation of the measurements	1.6×10^{-3}	3×10^{-3}
Step2: Prediction of particle states	5.8×10^{-5}	5.9×10^{-5}
Step3: Calculation of particle weights	7.2×10^{-4}	1.6×10^{-3}
Step4: Weights normalization	2.5×10^{-6}	2.4×10^{-6}
Step5: Resample	7.5×10^{-5}	7.3×10^{-5}

sensitive to the setting of initialization. The closer the initial value is to the true state of the tone at time $\kappa = 0$, the faster the convergence speed of the algorithm will be, and more particles will gather near the true state. A suitable initial value enables a more satisfactory tracking result. Therefore, when there are multiple tones, multiple particle filters could be performed by making the corresponding initial values as the initial states of different frequency bins. In other words, M filters are developed in M bins to work together to give the tracking results for multiple tones.

3.3. Comparison with SF-TBD**3.3.1. Comparison of SNR gain**

The SNR in measurements of MFC-TBD is higher than SF-TBD thanks to the coherent integration between multi-frame data. Ideally, the additional gain that the MFC-TBD method can provide is [12]:

$$G = 10 \log_{10} \left[W \left(\sum_{w=1}^W \frac{1}{w} \right)^{-1} \right] \quad (30)$$

A visual graph of (30) is shown in Fig. 4. It indicates that the

additional gain increases as the batch length increases. For example, an additional 3.9 dB gain can be provided when the batch length is 6. But excessively long batch lengths can reduce the real-time performance of the detector. In fact, the coherence length of the signal is limited. Especially in the harsh marine environment, the coherence time of tones may be only tens of seconds. In addition, the detection performance of the proposed method is also affected by the batch length (which will be analyzed in detail in section IV-3). Therefore, the batch length W should be set according to actual needs.

3.3.2. Comparison of computational cost

The computational costs of the SF-TBD and MFC-TBD methods are analyzed in this section. According to the general framework of PF in Algorithm 1, the computational cost is mainly determined by the iterative process (lines 3 to 9) at each time κ . Consider that the costs of generating a random number, evaluating special functions and resampling cannot easily be measured by the number of floating-point-operations. The time costs of SF-TBD and MFC-TBD algorithms are investigated through a computer with an Intel Core i5-1240P CPU 1.70 GHz and 16GB RAM, running Windows 11 Home.

The length of the simulated signal is 1000s. The frame length is set to 10 s and the number of particles is 500. 10,000 Monte Carlo trials are performed using MATLAB 2022b software. The averaged time cost of each step is listed in Table 1. The total running time of the MFC-TBD method is 0.048 s, which is about twice that of the SF-TBD method. For the MFC-TBD method, the experimental results show that the running time remains almost constant when the batch length is increased from 2 to 20 frames. In practice, the batch length is usually set to 4–8 frames. Therefore, the two methods are all achievable in real-time.

3.3.3. Comparison of detection performance

The result of TBD filter is posterior probability of tonal existence $P(e_k = 1 | Z_{1:k})$. If the probability is greater than a predetermined threshold η_{1k} , it can be declared that the tone has been detected. In mathematical terms, it is

$$\Lambda_{1k} = P(e_k = 1 | Z_{1:k}) \geq \eta_{1k} \quad (31)$$

where $Z_{1:k} = [Z_1, Z_2, \dots, Z_k]$ stands for the measurements during time 1 : k . According to the relationship between the posterior probabilities of tonal existence and nonexistence

$$P(e_k = 0 | Z_{1:k}) = 1 - P(e_k = 1 | Z_{1:k}) \quad (32)$$

the alternative test statistic can be defined as the ratio of the two posterior probabilities

$$\Lambda_{2k} = \frac{P(e_k = 1 | Z_{1:k})}{P(e_k = 0 | Z_{1:k})} \geq \eta_{2k} \quad (33)$$

Ignoring the effects of the tonal-birth probability P_b and tonal-death

probability P_d , the test statistic Λ_{2k} is derived [40] as

$$\Lambda_{2k} = \frac{P(e_k = 1|Z_{1:k})}{P(e_k = 0|Z_{1:k})} \quad (34)$$

$$\approx \frac{P(e_{k-1} = 1|Z_{1:k-1})}{P(e_{k-1} = 0|Z_{1:k-1})} \int l(Z_k|X_k, e_k) p(X_k|e_k = 1, e_{k-1} = 1, Z_{1:k-1}) dX_k$$

where $l(Z_k|X_k, e_k)$ denotes the likelihood ratio function, i.e. (16) in SF-TBD method and (26) in MFC-TBD method. $p(X_k|e_k = 1, e_{k-1} = 1, Z_{1:k-1})$ is the predictive probability density function. Transforming (34) from the state domain to the observation domain [41], we obtain

$$\Lambda_{2k} \approx \frac{P(e_{k-1} = 1|Z_{1:k-1})}{P(e_{k-1} = 0|Z_{1:k-1})} \int l(Z_k|y_k, e_k) p(y_k|e_k = 1, e_{k-1} = 1, Z_{1:k-1}) dy_k \quad (35)$$

where the observation y_k is the nonlinear function of target state X_k , i.e. $y_k = h(X_k)$. By using the set of particles $\{X_k^{(i)}\}_{i=1}^{N_p}$ and their corresponding weights to approximate the integration of y_k as a summation of discrete sample points, the posterior probability ratio test in (33) can be rewritten as

$$\Lambda_{2k} \approx \frac{P(e_{k-1} = 1|Z_{1:k-1})}{P(e_{k-1} = 0|Z_{1:k-1})} \sum_{i=1}^{N_p} w_k^i l(Z_k|y_k^{(i)}, e_k) \geq \eta_{2k} \quad (36)$$

In (36), $\frac{P(e_{k-1} = 1|Z_{1:k-1})}{P(e_{k-1} = 0|Z_{1:k-1})}$ is a constant at time k . Due to the resampling step in particle filtering, the predicted probability density of each particle is uniform and can be approximated as a constant. Therefore, the test based on (36) can be redefined as

$$\Lambda_{3k} = \sum_{i=1}^{N_p} l(Z_k|y_k^{(i)}, e_k) \geq \eta_{3k} \quad (37)$$

Generally, the analytical expression for the probability distribution of the test statistic in (37) is difficult to solve. Here, we make an approximation to this problem with the aim of using the trend of the detection performance of different methods derived from the approximation to deduce the detection performance determined by (31).

Define the sets Y_k and $\tilde{Y}_k$ as follows respectively:

$$Y_k = \left\{ y_k^{(i)} \left| \sum_{i=1}^{N_p} l(Z_k|y_k^{(i)}, e_k) > \eta_{3k} \right. \right\} \quad (38)$$

$$\tilde{Y}_k = \left\{ y_k \left| l(Z_k|y_k, e_k) > \frac{\eta_{3k}}{N_p} \right. \right\} \quad (39)$$

Comparing (38) and (39) shows that $\tilde{Y}_k$ is a subset of Y_k . This implies that if $l(Z_k|y_k, e_k) > \frac{\eta_{3k}}{N_p}$ holds, then $\sum_{i=1}^{N_p} l(Z_k|y_k^{(i)}, e_k) > \eta_{3k}$ must hold. So we can transform (37) into

$$\Lambda_{4k} = l(Z_k|y_k, e_k) \geq \eta_{4k} \quad (40)$$

where the threshold η_{4k} is given by $\eta_{4k} = \frac{\eta_{3k}}{N_p}$. Since the likelihood ratio

Table 3

Results of the false alarm probability indicator variable for SF-TBD and MFC-TBD methods.

Method	$\tilde{P}_f$					
SF-TBD	0.3684					
MFC-TBD	$W = 2$	$W = 4$	$W = 6$	$W = 8$	$W = 10$	$W = 12$
	0.4145	0.4502	0.4681	0.4803	0.4852	0.4937

$l(Z_k|y_k, e_k)$ is increasing with the increase of the observations y_k , (37) can be further transformed as:

$$\Lambda_{5k} = y_k \geq \eta_{5k} \quad (41)$$

From the above analysis, it can be shown that the comparison of the performance of the SF-TBD and MFC-TBD methods determined by (41) can to some extent illustrate the comparison results obtained from (31). In general, the likelihood ratio is theoretically equal to 1 when there is no target exist in the observed data [24]. Taking the threshold η_{4k} equals to 1 as an example, the detection performance of the two methods can be compared with the following analysis.

When a target is present in the received data, a detection probability indicator variable $\tilde{P}_d$ is defined as

$$\tilde{P}_d = \int_{\eta_{5k}}^{+\infty} p(y_k|e_k = 1) dy_k \quad (42)$$

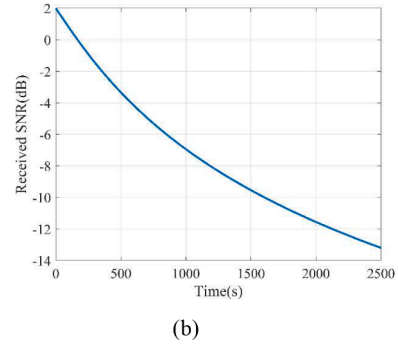
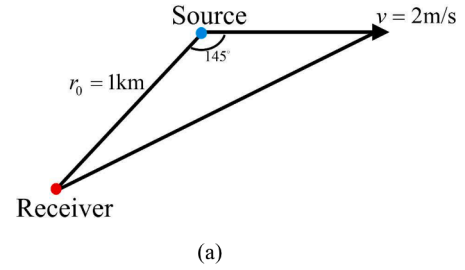


Fig. 5. The scenario where the source moves away from the receiver. (a) Source-receiver geometry (b) Received SNR.

Table 2

Results of the detection probability indicator variable for SF-TBD and MFC-TBD methods.

Input SNR (dB)	Method	$\tilde{P}_d$					
-10	SF-TBD	0.5569					
	MFC-TBD	$W = 2$	$W = 4$	$W = 6$	$W = 8$	$W = 10$	$W = 12$
		0.6321	0.7169	0.7856	0.8432	0.8887	0.9226
-7	SF-TBD	0.6767					
	MFC-TBD	$W = 2$	$W = 4$	$W = 6$	$W = 8$	$W = 10$	$W = 12$
		0.7678	0.8787	0.9414	0.9733	0.9881	0.9948
-4	SF-TBD	0.8145					
	MFC-TBD	$W = 2$	$W = 4$	$W = 6$	$W = 8$	$W = 10$	$W = 12$
		0.9084	0.9802	0.9962	0.9993	0.9999	1

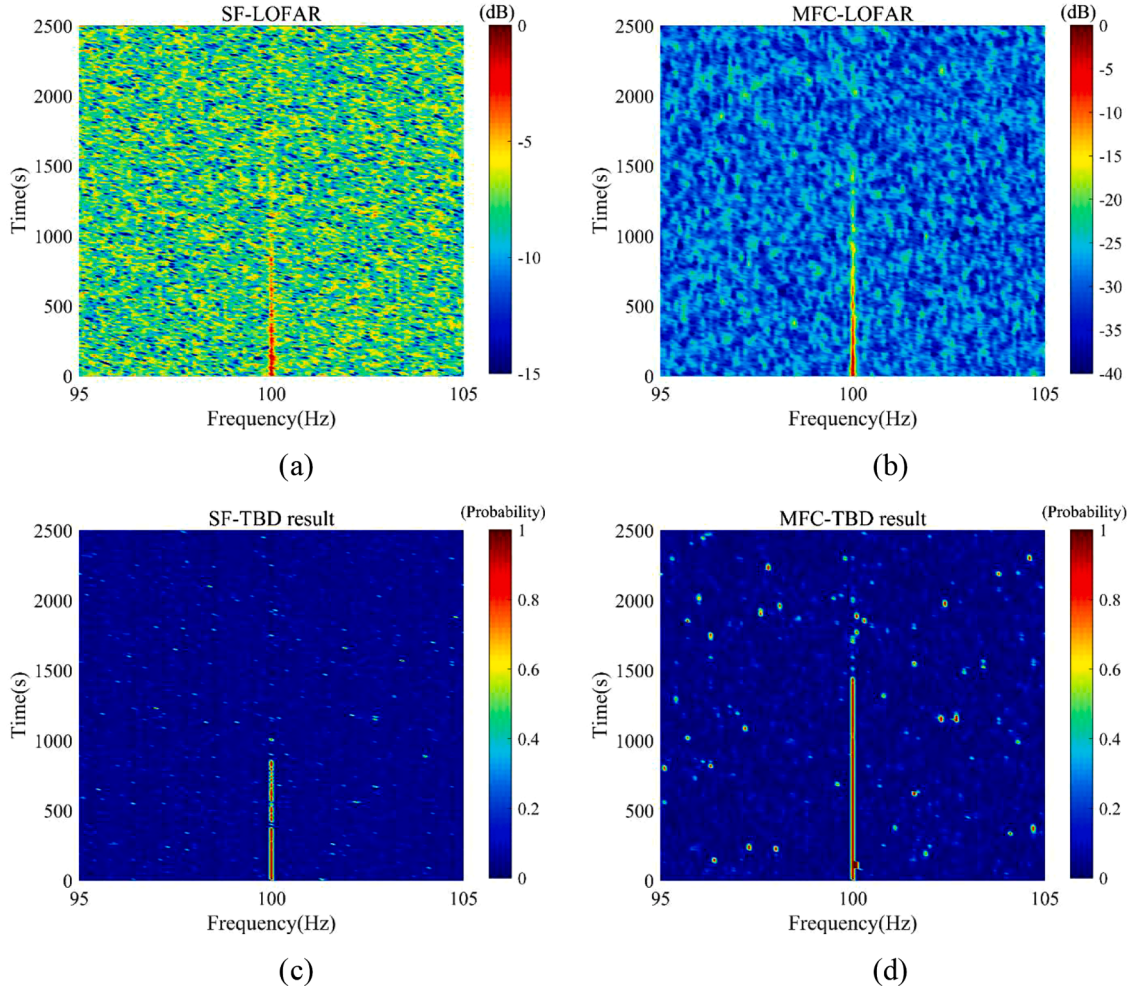


Fig. 6. Detection performance for a situation where the target is moving away from the receiver is depicted by (a) LOFAR of SF-TBD method, (b) LOFAR of MFC-TBD method, (c) SF-TBD method detection result, (d) MFC-TBD method detection result.

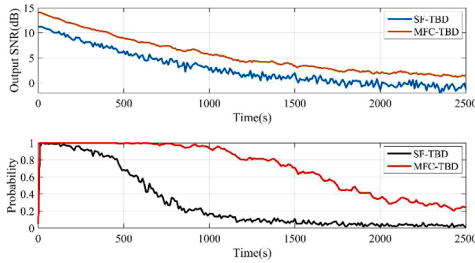


Fig. 7. Comparison of output SNRs in LOFARs and estimated tonal-existence probabilities in TBD results between SF-TBD and MFC-TBD methods.

$$\eta_{5k} = y_k |_{l(y_k|X_k, e_k) |_{\text{target present}} = 1} \quad (43)$$

and a false alarm probability indicator variable $\tilde{P}_f$ is defined as follows in the absence of a target

$$\tilde{P}_f = \int_{\eta_{5k}}^{+\infty} p(y_k | e_k = 0) dy_k \quad (44)$$

$$\eta_{5k} = y_k |_{l(y_k|X_k, e_k) |_{\text{not target}} = 1} \quad (45)$$

For the SF-TBD method, $p(y_k | e_k = 0)$, $p(y_k | e_k = 1)$ and $l(Z_k | y_k, e_k)$ are (14)–(16), respectively. And for the MFC-TBD method, they are (24)–

(26) respectively.

For the detection problem described in (31), a larger $\tilde{P}_d$ indicates a higher probability of detection and a larger $\tilde{P}_f$ indicates a higher probability of false alarm. To explore the performance of the SF-TBD and MFC-TBD methods, a simulated calculation is conducted on $\tilde{P}_d$ and $\tilde{P}_f$. The tonal frequency is set to 100.02 Hz and the sample frequency is 15 kHz. The data length of each frame is 10 s. For MFC-TBD method, the batch lengths are set as $W = 2, 4, 6, 8, 10, 12$, respectively. The results of $\tilde{P}_d$ at SNRs of -10 , -7 , and -4 dB are shown in Table 2. And the results of $\tilde{P}_f$ are shown in Table 3.

From the results in Tables 1 and 2, it can be concluded that $\tilde{P}_d$ of the MFC-TBD method is larger than that of the SF-TBD method at different SNR conditions, and it increases with the increase of batch length. However, the MFC-TBD method increases $\tilde{P}_f$ while increasing $\tilde{P}_d$, and $\tilde{P}_f$ also increases with batch length. This is mainly due to the fact that the operation of taking the maximum in MFC-TBD method destroys the original probability distribution of the noise. The proportion of larger measurements in the new probability distribution becomes larger, and the larger the batch length is, the more obvious this phenomenon is. At the same time, the proposed method integrates the information of multiple frames of a batch to obtain the processing gain, which is larger as the batch length increases, and thus a higher $\tilde{P}_d$ can be obtained. Therefore, the batch length should be carefully set in practical applications to expect a trade-off between the probability of detection and the probability of false alarms.

4. Performance analysis with synthetic data

In this section, the performance of the proposed algorithm is explored with synthetic data. For a more comprehensive evaluation of the algorithm performance, the performance of the multi-frame non-coherent TBD method (referred to as MFNC-TBD) [42] in specific scenarios is also included. For the simulation, the particle filter parameters are set according to [33].

4.1. Detection results for a moving target

The schematic of source motion is shown in Fig. 5(a). The time length of the simulated signal is 2500s and the sample frequency is 15 kHz. The data in each frame has 10 s and the batch length is set to 6 frames. The tonal frequency is set to 100.02 Hz. In this scenario, the Doppler frequency shift within a single batch is a maximum of 0.007 Hz, which is significantly less than $\Delta f = 0.1$ Hz. Thus the Doppler frequency shift is ignored, only the change in received SNR is considered.

As shown in Fig. 5(b), the received SNR decreases with time gradually, that is, from 2 to -13 dB. The detection and tracking results of an experiment show in Fig. 6. Fig. 7 displays the statistical results of 100 Monte Carlo experiments.

Fig. 6(a) and (b) represent LOFAR graphs of the measurements obtained by SF-TBD and MFC-TBD, respectively. Apparently, the SNR in MFC-LOFAR is higher and the tone presents a clearer trajectory. Fig. 6(c) and (d) show the results of tonal-existence probability estimated through the two TBD filters respectively. By comparison, it can be found that the detectable time length of MFC-TBD is about 600 s longer than that of SF-TBD and the detection performance of the former is steadier.

The output SNRs of the two methods estimated through 100 independent repeated experiments are averaged and plotted in the upper subgraph of Fig. 7. It can be seen that the MFC-TBD method can stably provide an additional gain of about 3 dB compared to the SF-TBD method. Similarly, the lower subgraph in Fig. 7 shows the statistical average results of the tonal-existence probabilities estimated by the two methods at 100 Hz. If the target is declared to be detected when the probability exceeds 0.8, the SF-TBD method can detect the target for only 450 s, while the MFC-TBD method can expand the detectable time length to 1210 s due to the vital improvement of output SNR. This means that the proposed MFC-TBD method may continuously track targets at longer distances in practice.

Based on the above results, we can conclude that the proposed method can improve the target SNR, thereby achieving a better detection performance at lower received SNR. Furthermore, compared with SF-TBD, the estimated tonal-existence probability of the MFC-TBD is higher and more stable. Multi frame information fusion enables it to obtain a better detection performance even in frames where the tone is too weak, reducing the risk of missing target.

4.2. Detection performance evaluation

Based on the test statistic described in (31), the detection performance of the SF-TBD and MFC-TBD methods is explored by simulation in this section. The detection threshold is obtained by the predetermined false alarm probability and 1000 Monte Carlo runs. To comprehensively compare the detection performance of different methods, define the following terms [26].

- 1) **Overall detection probability** is the fraction of Monte Carlo trials for which the tone is detected at any time.
- 2) **Instantaneous detection probability** is the total fraction of frames for which the tone is detected.
- 3) **Overall ROC** is the curve generated by plotting the overall detection probability as a function of the false alarm probability.
- 4) **Instantaneous ROC** is the curve generated by plotting the instantaneous detection probability as a function of the false alarm probability.

The performance of the SF-TBD and MFC-TBD is compared by simulating different scenarios with a single target under different SNR. Each scenario includes 100 data frames. The overall and instantaneous ROC curves of 100 experiments for -12 , -8 and -5 dB targets are shown in Fig. 8 and Fig. 9, respectively.

Figs. 8 and 9 indicate that for any given false alarm probability, the overall and instantaneous detection probabilities of MFC-TBD are higher than SF-TBD. In Fig. 8, when SNR is -5 dB, the overall detection probability of the MFC-TBD method is close to 1 for different false alarm probabilities, but that of the SF-TBD method can reach 1 only when the false alarm probability is close to 1. Similarly, in Fig. 9, when the instantaneous detection probability is 1, the false alarm probabilities required for MFC-TBD under different SNRs are 0.021, 0.005, and 0.001, respectively, while for SF-TBD method they are 0.322, 0.045, and 0.079, which are much larger than MFC-TBD. It can be seen that the overall detection probability is more stringent than the instantaneous detection probability, since the former requires the detector to fully detect a tone with a long duration.

Fig. 10(a) and (b) respectively show the overall and instantaneous detection probabilities of the two methods versus SNR under given false alarm probabilities of 0.1%, 0.5%, and 1%. As expected, the detection probability of MFC-TBD is higher than that of SF-TBD method under different SNRs.

In Fig. 10(a), the false alarm probability has little effect on the overall detection probability curve. Compared with the SF-TBD method, MFC-TBD can provide a gain of 3–4 dB robustly. This allows it to provide a higher detection probability under the same conditions. For example, when the signal-to-noise ratio is -5 dB, the overall detection probability of MFC-TBD can reach 1, while SF-TBD is only about 0.5.

In Fig. 10(b), there is a considerable gap between the instantaneous detection probability curves corresponding to different false alarm

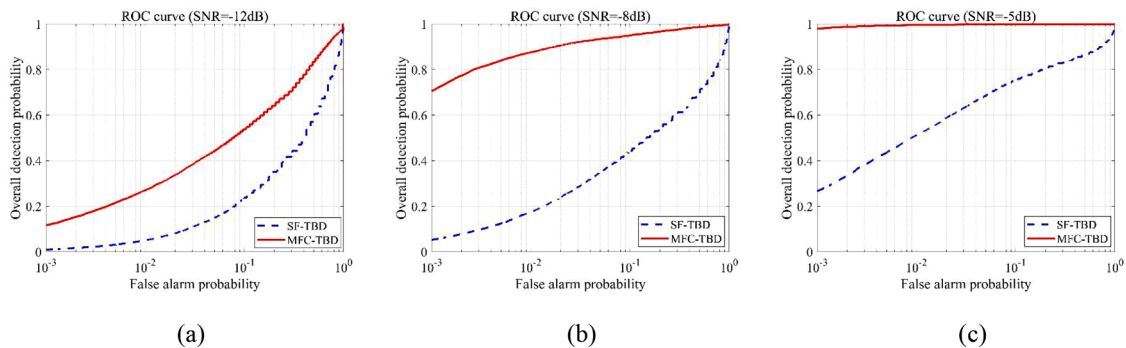


Fig. 8. Overall ROC curves of the SF-TBD and MFC-TBD at SNRs of (a) -12 (b) -8 (c) -5 dB.

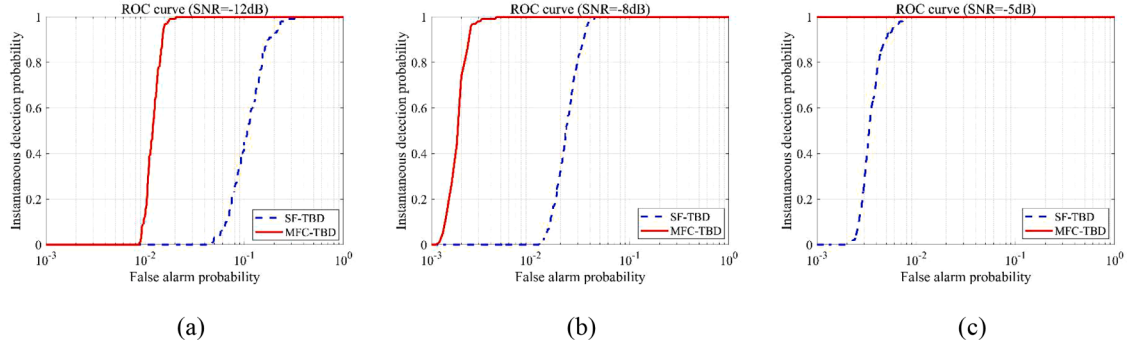


Fig. 9. Instantaneous ROC curves of the SF-TBD and MFC-TBD at SNRs of (a) -12 (b) -8 (c) -5 dB.

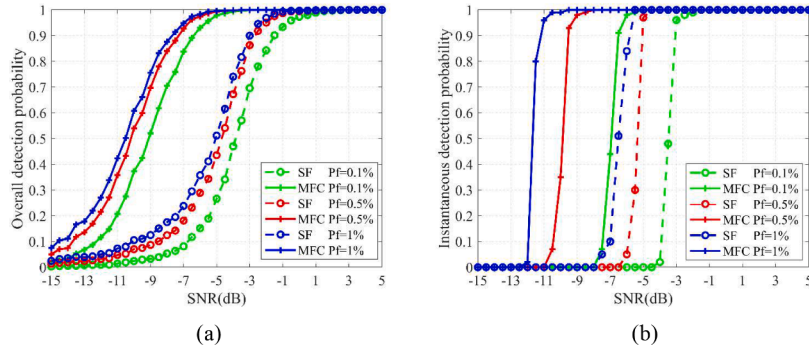


Fig. 10. (a) Overall detection probabilities and (b) Instantaneous detection probabilities of the SF-TBD (denoted as SF) and MFC-TBD (denoted as MFC) at false alarm probabilities of 0.1%, 0.5%, and 1%.

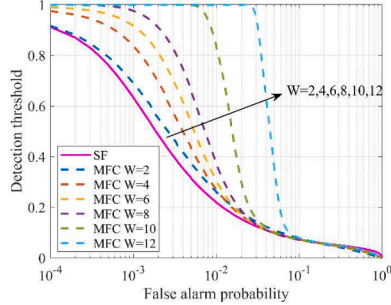


Fig. 11. Detection threshold curve for SF-TBD (denoted as SF) and MFC-TBD (denoted as MFC) with different batch length.

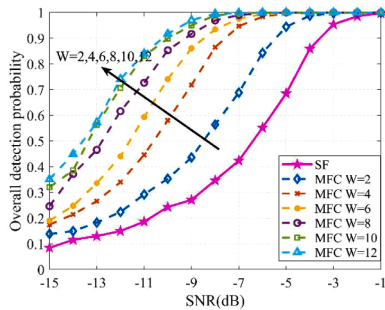


Fig. 12. Overall detection probabilities of SF-TBD (denoted as SF) the MFC-TBD (denoted as MFC) with different batch length at false alarm probability of 5%.

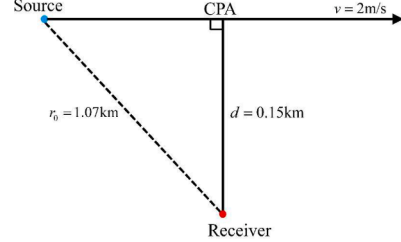


Fig. 13. Source-receiver geometry when a target passes through the receiver.

probabilities. The difference between the three curves for each method is approximately 1 dB and 2.5 dB, respectively. This is because the instantaneous detection probability represents the frame rate for which the target is detected, which makes it easier to reach a high level as the false alarm probability increases under the same SNR. As with the overall detection probability, for the same instantaneous detection probability, the SNR threshold of SF-TBD is 3–4 dB higher than that of MFC-TBD. The difference is that the measurement of instantaneous detection probability is often more tolerant of algorithms and can obtain a measure of high detection probability at lower SNRs.

4.3. Effect of the batch length

In this simulation, the impact of the batch length on the detection performance of the proposed MFC-TBD method is investigated. The simulation parameters remain unchanged. The relationship between the detection threshold and false alarm probability of the MFC-TBD method under different batch lengths is shown in Fig. 11. For comparison, the detection threshold curve of the SF-TBD method is also given.

In Fig. 11, it is undeniable that the detection thresholds of SF-TBD and MFC-TBD decrease with the increase of false alarm probability.

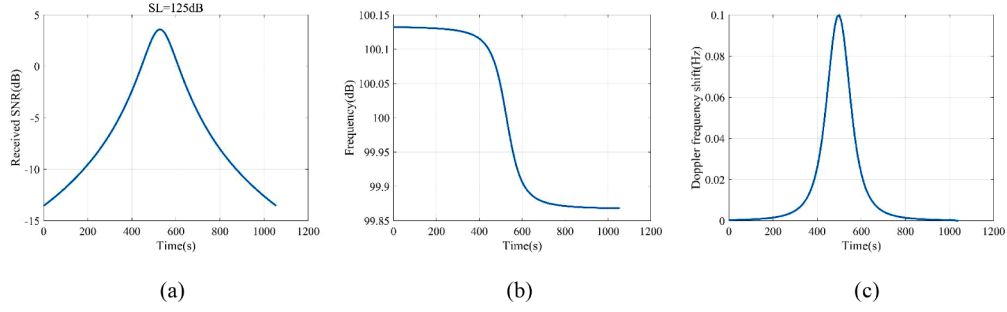


Fig. 14. (a) SNR of the received signal. (b) Frequency variation of the received signal. (c) Doppler frequency shift within a single batch at different time instant.

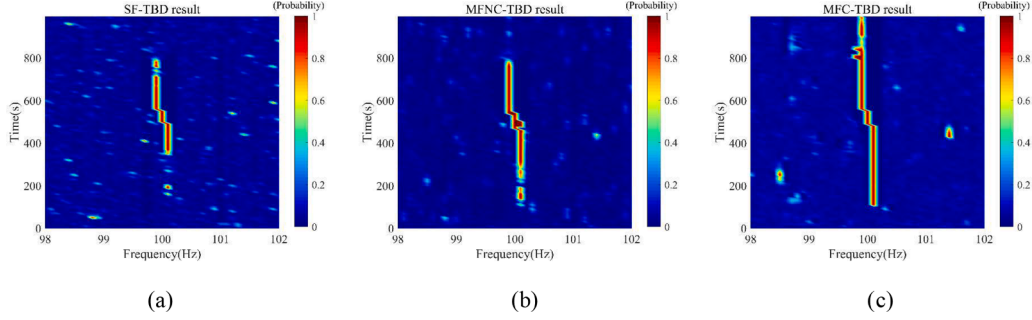


Fig. 15. Detection result of the (a) SF-TBD (b) MFNC-TBD and (c) MFC-TBD methods.

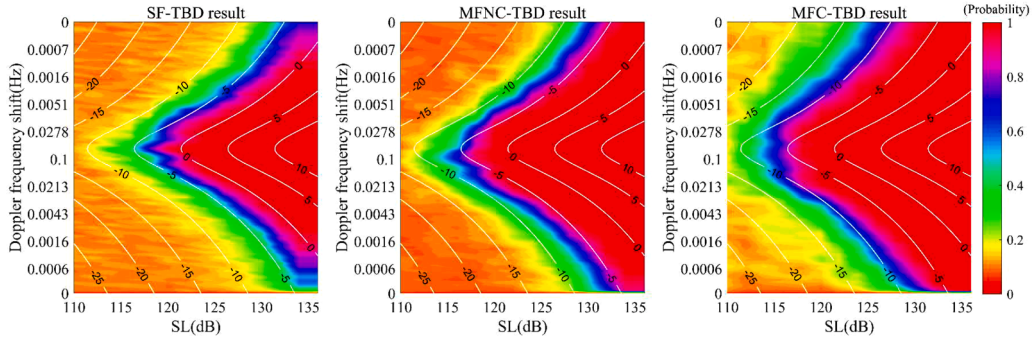


Fig. 16. Statistical average results on the tonal-existence probability at signal frequency corresponding to each Doppler frequency shift at different source levels (SLs) are depicted by (a) SF-TBD, (b) MFNC-TBD and (c) MFC-TBD methods.

When the false alarm probability is greater than 10%, the detection threshold curves of the two methods basically overlap. When the false alarm probability is less than 10%, as the batch length increases, the detection threshold curve of MFC-TBD gradually increases from near the threshold curve of SF-TBD. It seems that the latter is the lower limit of the former.

For MFC-TBD method, when the batch length are 10 and 12 frames, the maximum false alarm probabilities corresponding to a detection threshold of 1 are 0.4% and 2.3%, respectively. When the batch length are 2, 4, 6, and 8 frames, the corresponding maximum false alarm probabilities are less than 0.01% and increase sequentially. This means that under the same detection threshold, the false alarm probability of the MFC-TBD increases as the batch length increases. And it is higher than that of SF-TBD, which is more obvious when the batch length is greater than 10 frames.

To investigate the detection performance of the proposed method under different batch lengths, a comparison of overall detection probabilities of MFC-TBD with that of SF-TBD at false alarm probability of 5% is shown in Fig. 12.

In Fig. 12, the performance of MFC-TBD is improved with increasing

batch length. When the overall detection probability is 0.9, the input SNRs corresponding to MFC-TBD for different batch lengths are approximately -5.3 , -7.4 , -8.3 , -9.2 , -9.9 and -10.1 dB. For SF-TBD, it is -3.3 dB. It can be seen that the detection probability curve grows slower after the batch length is greater than 10 frames.

Based on the results of Figs. 11 and 12, we found that in order to improve the detection performance of the MFC-TBD method while keeping the false alarm probability within a tolerable range, it is best to set the batch length between 4 and 8 frames.

4.4. Simulation analysis under different Doppler frequency shifts

In this simulation, to more accurately assess the impact of Doppler frequency shift on algorithm performance, the scenario of a moving target passing through the receiver is simulated. The source-receiver geometry is shown in Fig. 13. Where CPA denotes the closest point of approach. The source frequency is 100 Hz. The frame length is 10 s and the batch length is set to 6.

If the source level (SL) is 125 dB, the received SNR calculated based on the passive sonar equation [33] is shown in Fig. 14(a). In Fig. 14(b),

the variation of signal frequency with time is plotted. And Fig. 14(c) illustrates the variation of Doppler frequency shift within a single batch. Obviously, the closer the target is to the CPA, the larger the Doppler frequency shift is. In other words, the greater the change rate of the signal frequency.

The simulated signal is processed by SF-TBD, MFNC-TBD and MFC-TBD methods respectively. The results for the tonal-existence probability are shown in Fig. 15. It shows that MFC-TBD outperforms SF-TBD and MFNC-TBD methods.

The analysis of Fig. 14 reveals that the Doppler frequency shift varies at different time instants. By manipulating the source level (SL), we can simulate scenarios with a constant Doppler frequency shift (i.e. a fixed time instant) while observing variations in the received SNR. Hence, to comprehensively investigate the performance of three algorithms under varying Doppler frequency shifts and different received SNRs, simulations are conducted for target motion scenarios at different SLs. For each SL, the detection results (akin to the presentation in Fig. 15) obtained from each Monte Carlo experiment are analyzed, and the tonal-existence probability at the signal frequency that is depicted in Fig. 14(b) is extracted. The statistical average results of the tonal-existence probability through 100 experiments are presented in each column of Fig. 16. To facilitate a more intuitive observation, using the correlation between Doppler frequency shift and time in Fig. 14(c), we have annotated the vertical axis in Fig. 16 with Doppler frequency shift values. The solid white lines in the graph represent contours of constant received SNR. In other words, points along the same contour exhibit equivalent SNR values.

As depicted in Fig. 16, the detection performance of the three TBD methods weakens with decreasing SNR at fixed Doppler frequency shift. Under a constant SNR, the detection performance of the SF-TBD method exhibits a slight decline as the Doppler frequency shift increases. Both the MFNC-TBD and MFC-TBD methods show a declining trend in performance, with the latter being more pronounced due to its heightened sensitivity to signal phase. Comparing the detection performance of the three methods, overall, $\text{MFC-TBD} > \text{MFNC-TBD} > \text{SF-TBD}$. Specifically, at a SNR of -10 dB, for smaller Doppler frequency shifts (< 0.02 Hz), the tonal-existence probabilities for the three methods are approximately 0.7, 0.4, and 0.2, respectively. Even with a Doppler frequency shift of 0.1 Hz, the MFC-TBD method exhibits superior performance compared to the other two methods.

For underwater targets with low frequency and low velocity, their Doppler frequency shift is typically small. The influence of Doppler effect on the algorithm performance can be effectively mitigated by judiciously selecting the window length based on (18). In summary, the proposed MFC-TBD method demonstrates superior detection performance. It can be applicable to most scenarios involving target motion.

5. Processing results of experimental data

In complex marine environment, strong fluctuations exist in the received signal due to the interference of multipath arrivals, irregularity

of boundary and the relative motion between the source and the receiver, etc. This fluctuation will make the signal processor more complex and degrade the detection performance of the detector. To verify the performance of the proposed algorithm in real marine environments, we conducted three trials in the South China Sea. The results indicate that the detection and tracking performance of MFC-TBD method is superior to SF-TBD method under different experimental scenarios.

5.1. First trial: detecting a tone from a stationary deterministic source

The schematic of the first trial is illustrated in Fig. 17. A continuous wave (CW) signal with a pure tone at 71 Hz was emitted by a stationary source. It consists of two CW pulses with a width of 600 s and an interval of 100 s. A 32-element line array was vertically placed to receive the signal propagating through the acoustic channel.

The signal received by one element is processed using segmented DFT and multi frame coherent integration, respectively. The processing results in the frequency range of 68–74 Hz are shown in Fig. 18. The segment length is 10 s and the batch length is 6 frames.

As shown in Fig. 18(a) and (b), the time length of the processed data is 1300s, which contains two CW pulses with durations of 0–600 s and 700–1300s, respectively. The intensity of the tone fluctuates with time due to the fluctuating effect of the acoustic signal. In particular, during the time period from 960 s to 1320s, the tonal energy drops rapidly and is almost submerged by noise. Because of the coherent accumulation of multiple data frames, the MFC-TBD method can reduce the level of strong background noise and obtain a clearer LOFAR than SF-TBD. The SNRs of the measurements obtained by the two methods were estimated in the analyzed frequency band and a comparative analysis is shown in Fig. 18(c). It can be seen that the SNR obtain by MFC-TBD is always greater than that of the SF-TBD, and its fluctuations are smaller. This creates a favorable condition for the subsequent detection and tracking using TBD filter.

Fig. 19 shows the results of the detection and tracking based on Fig. 18. For the first pulse, both methods detect the tone correctly, except that at first the SF-TBD method presents intermittent results due to the low SNR. As can be seen from Fig. 18 that the latter half of the second pulse is almost submerged in noise, so that the SF-TBD result only yields a weak trajectory. Surprisingly, the MFC-TBD method can detect the entire pulse perfectly. Furthermore, even if there is strong noise around 1130s, the detection of the weak tone during this time period is not affected. The tonal-existence probabilities estimated by the two methods are compared more visually in Fig. 19(c). The detection probability of the MFC-TBD method is higher and more stable than that of SF-TBD. For example, the detection probability of SF-TBD is only about 0.2 in the 960s–1320s, but MFC-TBD can reach about 0.8. In addition, at times when the detection probability of SF-TBD is almost zero, MFC-TBD achieves a probability of around 0.6.

The processing results of a stationary source strongly suggest that the proposed method is effective in reducing the effect of marine acoustic channel undulations on tonal detection.

5.2. Second trial: detecting a tone from a moving deterministic source

In actual combat scenarios, it is of great strategic importance to be able to detect targets coming from a distance at an early stage. As in the first trial, the experimental configuration shown in Fig. 17 is considered again in the second trial. The vessel carrying a towed source was driven to move from a distance towards a direction close to the receiver. With the receiver as the coordinate origin, the relative trajectory of the moving source is shown in Fig. 20. The source was essentially moving in a straight line along the radial direction of the receiver. Within the 2500s of observation, the initial and final distances between the source and the receiver were 8.49 and 1.85 km, respectively.

To verify the detection performance of the algorithm under poor

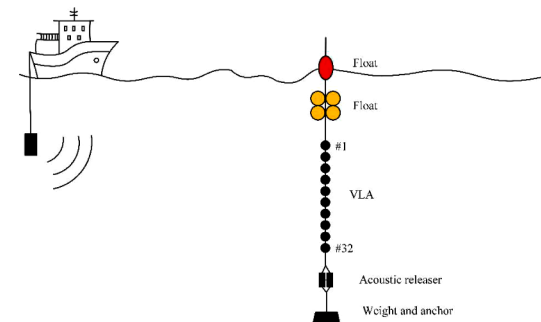


Fig. 17. Schematic for the first trial.

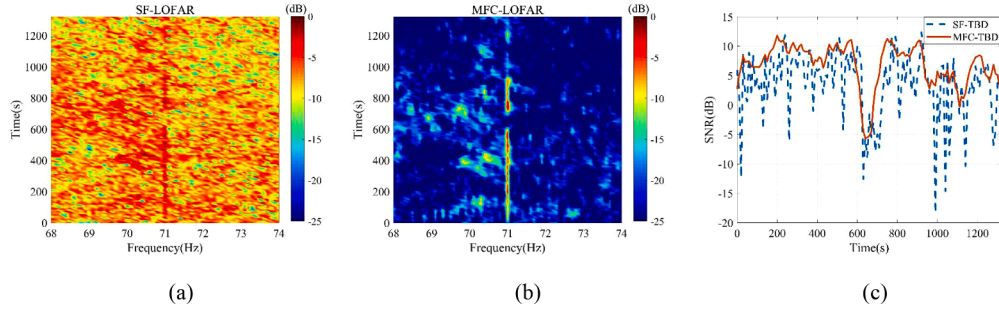


Fig. 18. Data from a stationary deterministic source is depicted by (a) LOFAR of SF-TBD method, (b) LOFAR of MFC-TBD method and (c) SNR comparison of measurements in SF-TBD and MFC-TBD.

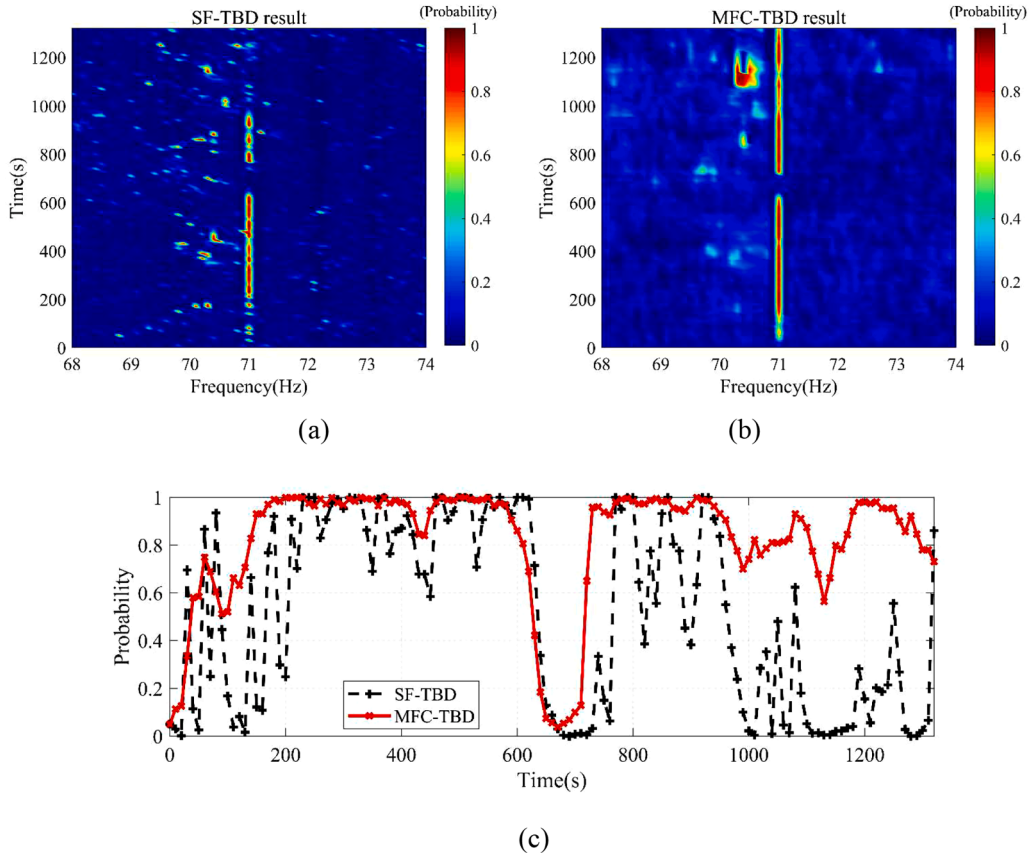


Fig. 19. Detections results of a stationary deterministic source are depicted by (a) SF-TBD method detection result (b) MFC-TBD method detection result and (c) detection results comparison of the two methods.

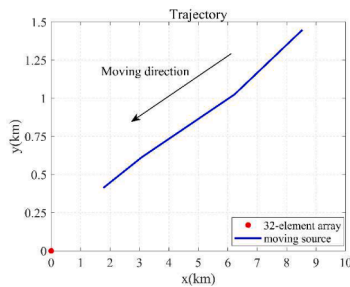


Fig. 20. Relative trajectory of the moving source.

SNR, real noise data collected at sea for the same time length was added to the original received data. The segment length and batch length were set to 10 s and 8 frames respectively. The LOFARs and the estimation results of tonal-existence probability obtained by using the SF-TBD and MFC-TBD methods are shown in Fig. 21 and Fig. 22 respectively.

As shown in Fig. 21(a) and (b), the intensity of the tone increases gradually with time. The first pulse is the weakest and is almost buried in noise. There is an overall increasing trend in the SNR of the measurements in Fig. 21(c) for both methods. This is due to the fact that the sound propagation loss in the water decreases as the distance between the source and receiver decreases. Compared with SF-TBD, the SNR curve for the MFC-TBD is higher and smoother.

In Fig. 22(a), the SF-TBD detects the last two pulses correctly. For the first pulse, however, there are only a few sporadic detection points. In contrast, as shown in Fig. 22(b), the MFC-TBD not only detects all pulses correctly, but also obtains a more stable estimate of tonal-

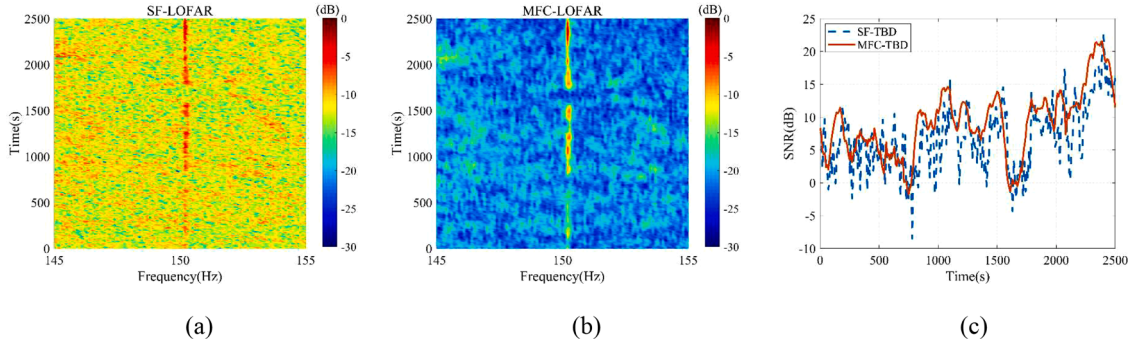


Fig. 21. Data from a moving deterministic source is depicted by (a) LOFAR of SF-TBD method, (b) LOFAR of MFC-TBD method and (c) SNR comparison of measurements in SF-TBD and MFC-TBD.

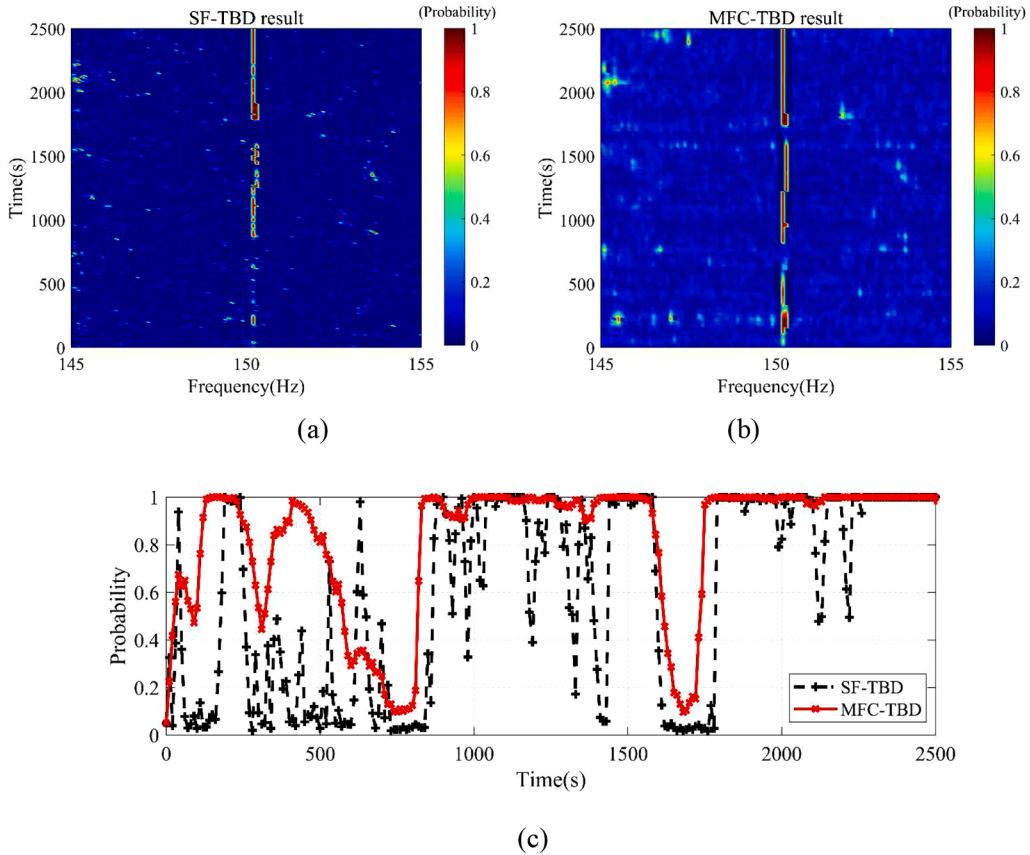


Fig. 22. Detections results of a moving deterministic source are depicted by (a) SF-TBD method etection result (b) MFC-TBD method detection result and (c) detection results comparison of the two methods.

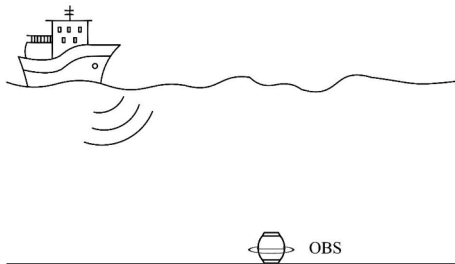


Fig. 23. Schematic for the third trial.

existence probability. This can be further observed in Fig. 22(c).

We can conclude that when detecting a moving target, the MFC-TBD is able to detect the tone at a longer distance earlier and more robustly.

5.3. Third trial: detecting tones from a surface vessel

In the third trial, we use an ocean-bottom seismometers (OBS) to collect the data radiated from a surface vessel. It can be viewed as an acoustic vector hydrophone. The configuration of the trial is shown in Fig. 23. The observation time was 4100 s. The segment length was set to 15 s and the batch length was 6 frames. The length of the protection window and reference window were set to 1 s and 5 s, respectively. The processing results for the sound pressure component of the OBS in the frequency range 13–24 Hz are shown in Figs. 24 and 25.

As can be seen in Fig. 24, there are eight tones whose frequencies

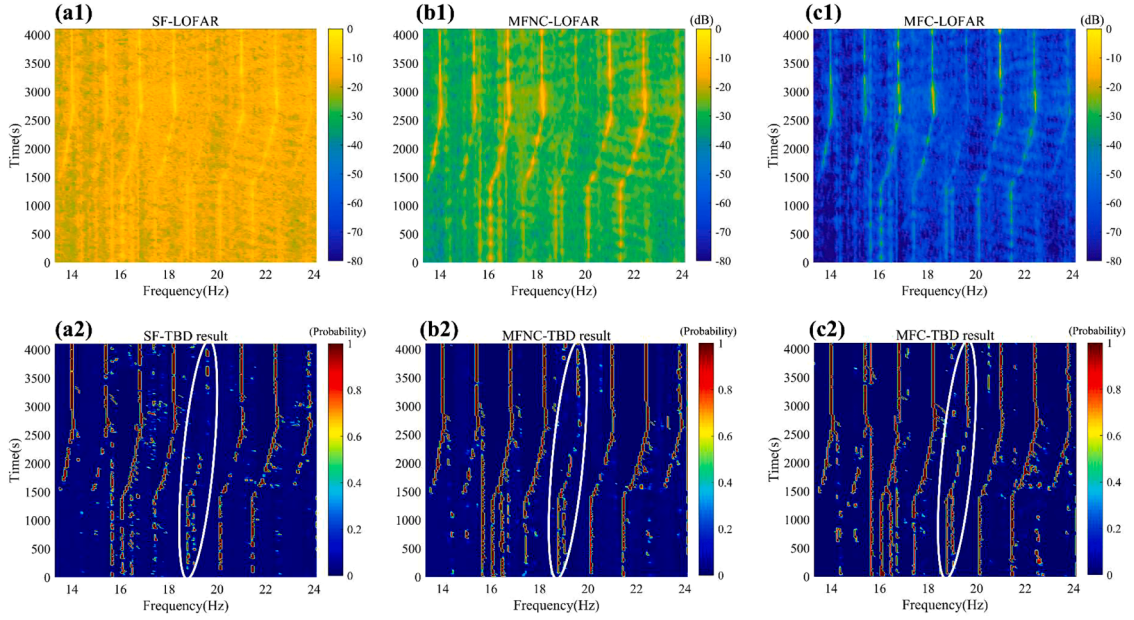


Fig. 24. Processing results of the real data from a surface vessel are depicted by (a) SF-TBD method, (b) MFNC-TBD method and (c) MFC-TBD method.

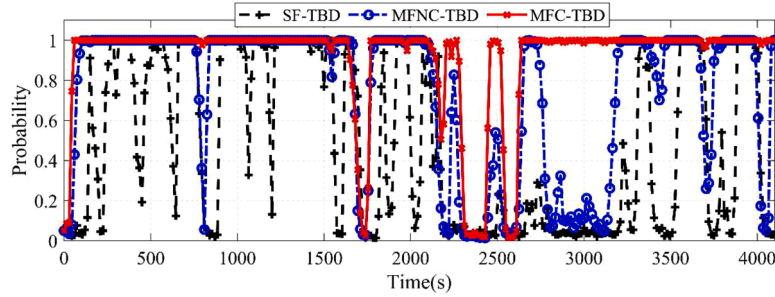


Fig. 25. Detection results comparison of the three methods for the 5th tone from the target vessel.

change dramatically with time due to the relative motion of the surface vessel and the receiver. They have a consistent frequency variation pattern and the frequency interval between two adjacent tones is about 1.4 Hz. In addition to this, the OBS also received some tones radiated from unknown targets, whose frequencies remain basically constant during the observation time, such as 15.67 Hz, 16.73 Hz and 19.07 Hz. By integrating tonal information over multiple frames, a reduction in the background noise level can be observed in the LOFAR of the MFNC-TBD and MFC-TBD methods. Furthermore, MFC-TBD method demonstrates a distinct superiority. Comparison of the detection results reveals that the SF-TBD detector misses a large number of potential true tones due to the signal fluctuations. MFNC-TBD and MFC-TBD detectors can clearly detect most of the tones and retain their frequency characteristics. Further, during multiple time intervals where the tonal frequency remains approximately constant, the probability of missed detection for the MFC-TBD method is significantly lower than that of the MFNC-TBD method.

Fig. 25 compares the detection results of the three methods for the 5th tone of the target vessel in the frequency range of 18.7–19.7 Hz. The detection probability curve of SF-TBD shows large fluctuations. To make matters worse, the target is never detected during 2200–3250 s. On the contrary, the curve of MFC-TBD method exhibits significantly lower fluctuations. However, it also fails to detect the target during the 2800–3100 s. Except for the period between 2300s and 2600s, the existence probability of the MFC-TBD method is nearly 1 at every time instant. Furthermore, even during this time interval, the detection

performance of the MFC-TBD method remains optimal.

By processing and analyzing the received data from a moving surface vessel, we find that even in complex marine environment, the proposed method can provide a more robust and higher tonal-existence probability at a lower SNR.

6. Conclusion

In this study, a multi-frame coherent track-before-detect method is proposed to achieve robust detection and tracking performance for weak tones in passive sonar. The target SNR is improved through multi-frame coherent integration based on the batch processing technique. The likelihood ratio function, which is more sensitive to tonal information, is integrated into the TBD filtering process. Therefore, compared to the SF-TBD method, the proposed method can improve the detection probability of low SNR targets. Simulation analysis shows that the MFC-TBD method can provide a gain of about 3–5 dB compared to the SF-TBD method at the same false alarm probability. The real data processing results in complex marine environments also indicate that the MFC-TBD method can effectively reduce the impact of fluctuations on tonal detection, and obtain more stable and higher estimates of tonal-existence probability. This means that it is suitable for detecting weak targets with low-frequency and low-velocity in real underwater situations.

Notably, the received signal may exhibit a significant Doppler frequency shift when the source frequency and target velocity are relatively

high. In such circumstances, the performance of the proposed algorithm may experience degradation or even failure. Therefore, in future work, the authors may consider investigating more robust methods under the influence of severe Doppler effects to extend the application scenarios of multi-frame processing.

CRediT authorship contribution statement

Liu Zhang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Shengchun Piao:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Junyuan Guo:** Writing – review & editing, Supervision, Formal analysis. **Xiaohan Wang:** Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix

According to (10)–(21), $Q_{B_k}(l, q) \sim CN(Q_{S_k}(l, q), 2\sigma_{Q_G}^2)$, where CN represents the complex Gaussian distribution. Define that

$$\tilde{Z}_k(l, q) = |Q_{B_k}(l, q)|^2 \quad (46)$$

Then the measurement for MFC-TBD method is $Z_k^M(l) = \max_q \tilde{Z}_k(l, q)$.

The distributions of $Z_k(l)$ when $l = l_{0k}$ and $l \neq l_{0k}$ are discussed in the following.

(1) $l \neq l_{0k}$: $\tilde{Z}_k(l, q_i)$, $i = 1, 2, \dots, W$ follows a chi-square distribution with two degrees of freedom and they are independent

$$p_{\tilde{Z}_k(l, q_i)}(x) = \frac{1}{2\sigma_{Q_G}^2} \exp\left\{-\frac{x}{2\sigma_{Q_G}^2}\right\}, i = 1, 2, \dots, W \quad (47)$$

Then the cumulative distribute function of $Z_k^M(l)|_{l \neq l_{0k}}$

$$F_{Z_k^M(l)|_{l \neq l_{0k}}}(x) = \left[\int p_{\tilde{Z}_k(l, q_i)}(x) dx \right]^W = \left[1 - \exp\left\{-\frac{x}{2\sigma_{Q_G}^2}\right\} \right]^W \quad (48)$$

Thus, the probability density function of $Z_k^M(l)$ when $e_k = 0$ is

$$p(Z_k^M(l)|e_k=0) = \frac{W}{2\sigma_{Q_G}^2} \left[1 - \exp\left\{-\frac{Z_k^M(l)}{2\sigma_{Q_G}^2}\right\} \right]^{W-1} \exp\left\{-\frac{Z_k^M(l)}{2\sigma_{Q_G}^2}\right\} \quad (49)$$

(2) $l = l_{0k}$ 时: $\tilde{Z}_k(l_{0k}, q_i)$, $i = 1, 2, \dots, W$, $i \neq l_{0k}$ follow a chi-square distribution with two degrees of freedom and they are independent. $\tilde{Z}_k(l_{0k}, q_{l_{0k}})$ follows a noncentral chi-square distribution with two degrees of freedom. That is

$$p_{\tilde{Z}_k(l_{0k}, q_i)}(x) = \frac{1}{2\sigma_{Q_G}^2} \exp\left\{-\frac{x}{2\sigma_{Q_G}^2}\right\}, i = 1, 2, \dots, W, i \neq l_{0k} \quad (50)$$

$$p_{\tilde{Z}_k(l_{0k}, q_{l_{0k}})}(x) = \frac{1}{2\sigma_{Q_G}^2} I_0\left(\frac{\sqrt{|Q_{S_k}(l_{0k}, q_{l_{0k}})|^2 x}}{\sigma_{Q_G}^2}\right) \exp\left\{-\frac{|Q_{S_k}(l_{0k}, q_{l_{0k}})|^2 + x}{2\sigma_{Q_G}^2}\right\} \quad (51)$$

where $I_0(\cdot)$ is the modified Bessel function of the first kind of order 0.

Then the cumulative distribute function of $Z_k^M(l)|_{l=l_{0k}}$ is

$$\begin{aligned}
F_{Z_k^M(l)}(x) &= [F_{Z_k(l_{0k}, q_l)}(x)]^{W-1} F_{Z_k(l_{0k}, q_{l_{0k}})}(x) \\
&= \left[\int_{-\infty}^x P_{Z_k(l_{0k}, q_l)}(x) dx \right]^{W-1} \int_{-\infty}^x P_{Z_k(l_{0k}, q_{l_{0k}})}(x) dx \\
&= \left[1 - \exp\left\{-\frac{x}{2\sigma_{QG}^2}\right\} \right]^{W-1} \int_0^x P_{Z_k(l_{0k}, q_{l_{0k}})}(x) dx
\end{aligned} \tag{52}$$

Thus, the probability density function of $Z_k^M(l)$ when $e_k = 1$ is

$$\begin{aligned}
p(Z_k^M(l)|e_k = 1) &= \left[1 - \exp\left\{-\frac{Z_k^M(l)}{2\sigma_{QG}^2}\right\} \right]^{W-1} \frac{1}{2\sigma_{QG}^2} I_0\left(\frac{\sqrt{|Q_{S_k}(l, q_l)|^2 Z_k^M(l)}}{\sigma_{QG}^2}\right) \exp\left\{-\frac{|Q_{S_k}(l, q_l)|^2 + Z_k^M(l)}{2\sigma_{QG}^2}\right\} \\
&\quad + \frac{(W-1)}{2\sigma_{QG}^2} \left[1 - \exp\left\{-\frac{Z_k^M(l)}{2\sigma_{QG}^2}\right\} \right]^{W-2} \exp\left\{-\frac{Z_k^M(l)}{2\sigma_{QG}^2}\right\} \int_0^{Z_k^M(l)} P_{Z_k(l_{0k}, q_{l_{0k}})}(x) dx
\end{aligned} \tag{53}$$

References

- [1] Y. Song, J. Liu, L. Cao, et al., Robust passive underwater acoustic detection method for propeller, *Appl. Acoust.* 148 (2019) 151–161.
- [2] L. Ma, T.A. Gulliver, A. Zhao, et al., Underwater broadband source detection using an acoustic vector sensor with an adaptive passive matched filter, *Appl. Acoust.* 148 (2019) 162–174.
- [3] A. Trabattoni, G. Barriuel, R. Dréo, et al., Ship detection and tracking from single ocean-bottom seismic and hydroacoustic stations, *J. Acoust. Soc. Am.* 153 (1) (2023) 260–273.
- [4] M.C. Özden, A.Y. Gürkan, Y.A. Özden, et al., Underwater radiated noise prediction for a submarine propeller in different flow conditions, *Ocean Eng.* 126 (2016) 488–500.
- [5] B. Zhang, Y. Xiang, P. He, et al., Study on prediction methods and characteristics of ship underwater radiated noise within full frequency, *Ocean Eng.* 174 (2019) 61–70.
- [6] Y. Hao, L. Qiu, C. Chi, et al., Sparsity-inducing frequency-domain adaptive line enhancer for unmanned underwater vehicle sonar, *Appl. Acoust.* 173 (2021) 107689.
- [7] P. Welch, The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms, *IEEE Trans. Audio Electroacoust.* 15 (2) (1967) 70–73.
- [8] R.A. Wagstaff, The AWSUM filter: a 20-dB gain fluctuation-based processor, *IEEE J. Ocean. Eng.* 22 (1) (1997) 110–118.
- [9] C.R. Wan, J.T. Goh, H.T. Chee, Optimal tonal detectors based on the power spectrum, *IEEE J. Ocean. Eng.* 25 (4) (2000) 540–552.
- [10] Q. Wang, G. Shi, C. Wan, Analysis and design of an optimum detector for weak sinusoidal signals, in: *Proceedings of the 6th International Conference on Signal Processing 2*, IEEE, 2002, pp. 1608–1611.
- [11] Q. Wang, C.R. Wan, A novel CFAR tonal detector using phase compensation, *IEEE J. Ocean. Eng.* 30 (4) (2005) 900–911.
- [12] H. Lan, P.R. White, N. Li, et al., Coherently averaged power spectral estimate for signal detection, *Signal Process.* 169 (2020) 107414.
- [13] Q. Xie, C. Chi, S. Jin, et al., Underwater tone detection with robust coherently-averaged power processor, *J. Mar. Sci. Eng.* 10 (10) (2022) 1505.
- [14] Z. Zhao, Q. Li, Z. Xia, et al., A single-hydrophone coherent-processing method for line-spectrum enhancement, *Remote Sens.* 15 (3) (2023) 659.
- [15] T.A. Lampert, S.E.M. O'Keefe, A survey of spectrogram track detection algorithms, *Appl. Acoust.* 71 (2) (2010) 87–100.
- [16] R.L. Streit, R.F. Barrett, Frequency line tracking using hidden Markov models, *IEEE Trans. Acoust.* 38 (4) (1990) 586–598.
- [17] S. Paris, C. Jauffret, A new tracker for multiple frequency line, in: *Proceedings of the IEEE Aerospace Conference 4*, IEEE, 2001 (Cat. No. 01TH8542) 4/1771–4/1782 vol. 4.
- [18] S. Paris, C. Jauffret, Frequency line tracking using HMM-based schemes, *IEEE Trans. Aerosp. Electron. Syst.* 39 (2) (2003) 439–449.
- [19] X. Xie, R.J. Evans, Multiple target tracking and multiple frequency line tracking using hidden Markov models, *IEEE Trans. Signal Process.* 39 (12) (1991) 2659–2676.
- [20] X. Xie, R.J. Evans, Multiple frequency line tracking with hidden Markov models-further results, *IEEE Trans. Signal Process.* 41 (1) (1993) 334.
- [21] G. Pulford, K. Tyson, A high performance 1-D hidden Markov model tracker for passive sonar systems, in: *Proceedings of the UDT Europe*, 2015.
- [22] K. Ma, G. Luo, J. Cheng, et al., Line spectrum extraction method based on hidden Markov model, *Electron. Lett.* 57 (12) (2021) 486–488.
- [23] M.S. Arulampalam, S. Maskell, N. Gordon, et al., A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking, *IEEE Trans. Signal Process.* 50 (2) (2002) 174–188.
- [24] D.J. Salmond, H. Birch, A particle filter for track-before-detect, in: *Proceedings of the American Control Conference 5*, IEEE, 2001, pp. 3755–3760 (Cat. No. 01CH37148).
- [25] M.G. Rutten, N.J. Gordon, S. Maskell, Recursive track-before-detect with target amplitude fluctuations, in: *Proceedings of the IEE Radar, Sonar and Navigation 152*, 2005, pp. 345–352.
- [26] S.J. Davey, M.G. Rutten, B. Cheung, A comparison of detection performance for several track-before-detect algorithms, *EURASIP J. Adv. Signal Process.* 2008 (2007) 1–10.
- [27] P. Zhang, J. Yan, L. Ma, et al., An IPDA based target existence assisted Bayesian detector for target tracking in clutter, *Signal Process.* 205 (2023) 108895.
- [28] J.V. Candy, Broadband processing in a noisy shallow ocean environment: a particle filtering approach, *IEEE J. Ocean. Eng.* 41 (4) (2016) 794–809.
- [29] Z. Liang, X. Ma, X. Dai, Robust tracking of moving sound source using scaled unscented particle filter, *Appl. Acoust.* 69 (8) (2008) 673–680.
- [30] Z.H. Michalopoulou, R. Jain, Particle filtering for arrival time tracking in space and source localization, *J. Acoust. Soc. Am.* 132 (5) (2012) 3041–3052.
- [31] W. Yi, L. Fu, Á.F. García-Fernández, et al., Particle filtering based track-before-detect method for passive array sonar systems, *Signal Process.* 165 (2019) 303–314.
- [32] T. Northardt, S.C. Nardone, Track-before-detect bearings-only localization performance in complex passive sonar scenarios: a case study, *IEEE J. Ocean. Eng.* 44 (2) (2018) 482–491.
- [33] D. Zhang, L. Gao, D. Sun, et al., Soft-decision detection of weak tonals for passive sonar using track-before-detect method, *Appl. Acoust.* 188 (2022) 108549.
- [34] M. Bruno, Bayesian methods for multispect target tracking in image sequences, *IEEE Trans. Signal Process.* 52 (7) (2004) 1848–1861.
- [35] W. Yi, M.R. Morelande, L. Kong, et al., An efficient multi-frame track-before-detect algorithm for multi-target tracking, *IEEE J. Sel. Top. Signal Process.* 7 (3) (2013) 421–434.
- [36] J. Wang, Y. Wei, T. Kirubarajan, et al., An efficient recursive multiframe track-before-detect algorithm, *IEEE Trans. Aerosp. Electron. Syst.* 99 (2018) 1 PP.
- [37] J. Wang, W. Yi, R. Hoseinnezhad, et al., An agile multi-frame detection method for targets with time-varying existence, *Signal Process.* 165 (2019) 133–143.
- [38] W.R. Blanding, P.K. Willett, Y. Bar-Shalom, et al., Directed subspace search ML-PDA with application to active sonar tracking, *IEEE Trans. Aerosp. Electron. Syst.* 44 (1) (2008) 201–216.
- [39] Q. Peng, W. Li, L. Kong, Multi-frame track-before-detect algorithm for passive sonar system, in: *Proceedings of the International Conference on Control, Automation and Information Sciences (ICCAIS)*, 2019.
- [40] C. Xi, Z. Cai, J. Yuan, A method of determining detection threshold for particle filter track-before-detect in passive sonar, *Acta Electron. Sin.* 50 (7) (2022) 1722.
- [41] S.Z. Xia, H.W. Liu, A method of determining detection threshold for bayesian track-before-detection in white complex Gaussian noise, *J. Electron. Inf. Technol.* 35 (3) (2013) 524–531.
- [42] L. Zhang, S. Piao, J. Guo, et al., Multi-frame non-coherent track-before-detect method for weak tones in passive sonar, in: *Proceedings of the International Conference on Frontiers of Ocean Science and Technology (ICFOST)*, Changsha, China, 2023 (in press).